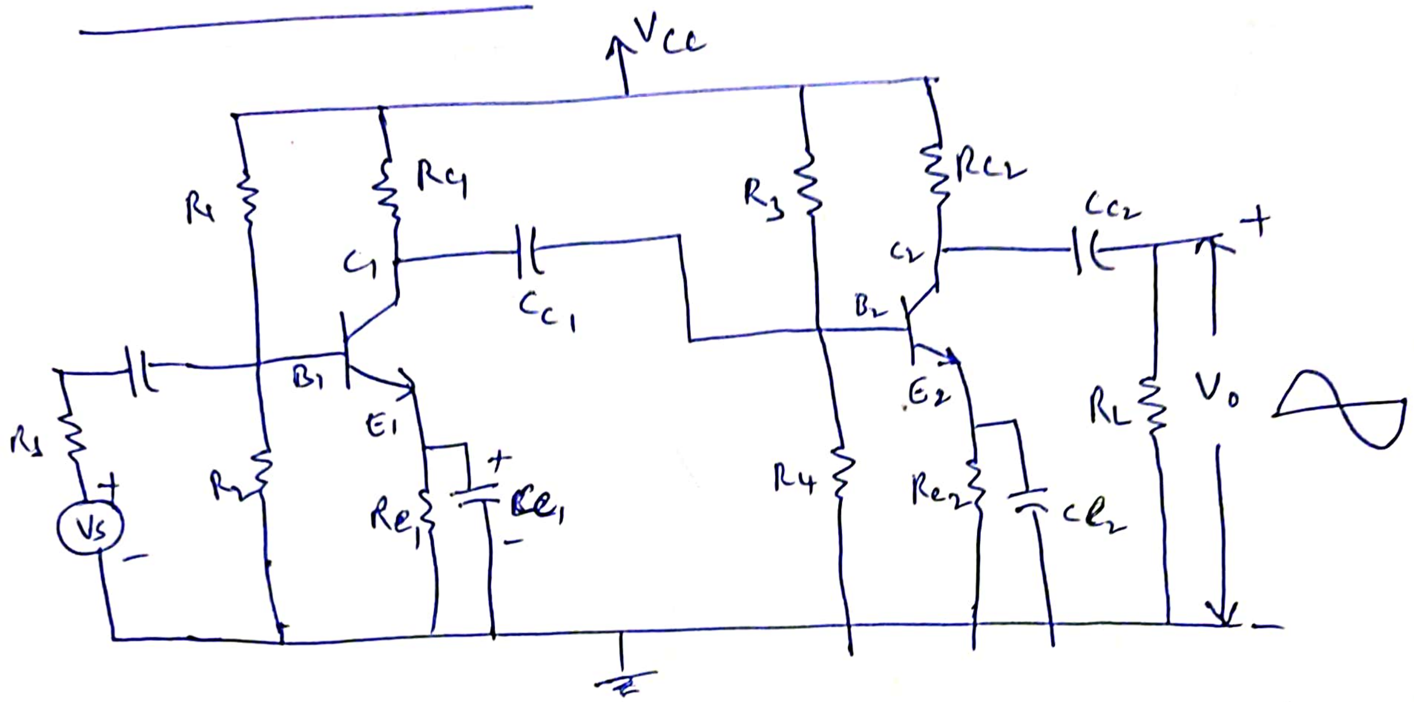
1. (ii): Multistage Amplifiers

RC coupled Amplifier



⇒ Two stage RC coupled amplifier which consists of two single stage CE amplifiers.

Construction :- (1) R_1, R_2, R_{C_1} and R_{E_1} along with Vcc are used to provide self bias and temperature stabilisation for transistor.

(2) The coupling capacitor C_{C_1} which connects the o/p of the first stage to base (i/p) of second stage. It allows only ac signals at the end of one stage to appear at the i/p of next and blocking the dc voltages and currents.

(3) The bypass capacitor (C_e) provides a low impedance path to the ac signal and prevents loss of amplification due to negative feedback. It provides high resistance for dc signal and does not allow any dc signal through it.

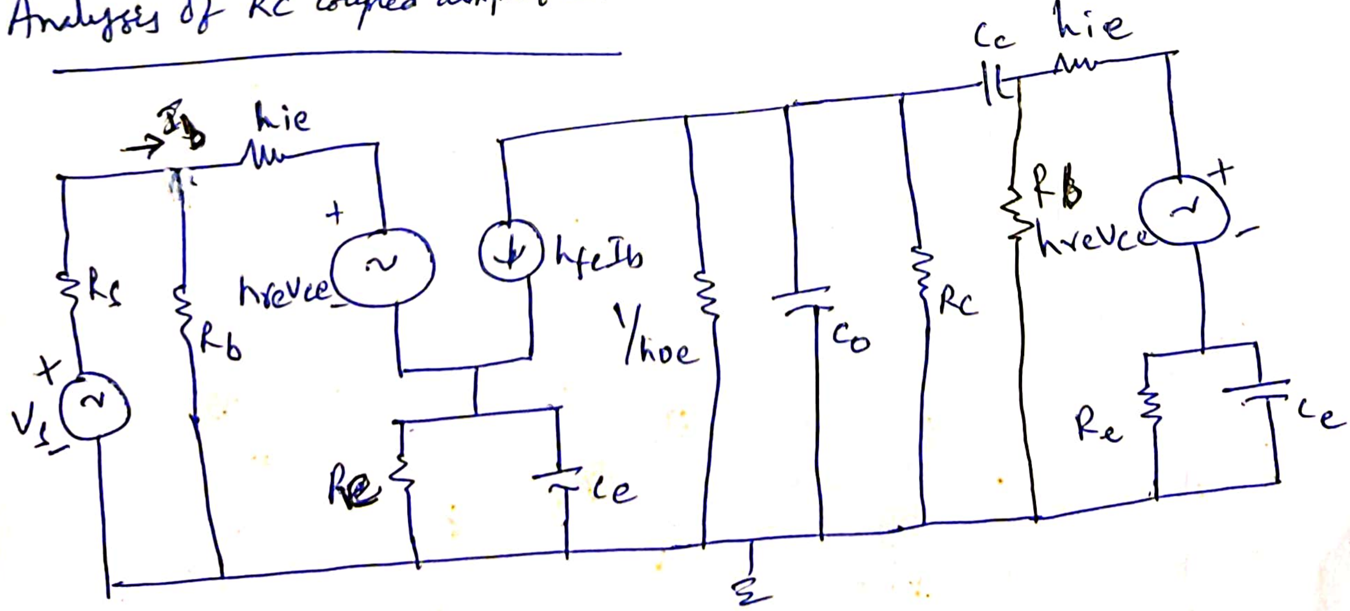
Operation :- (1) When the ac signal is applied to the base of first stage, it is amplified (because of CE connection). The amplified signal is given to the base of second stage through C_C.

(2) The signal at the base of second stage is further amplified and its phase is further reversed. The o/p of the second stage is coupled by C_C to the load Resistor R_L.

(3) In This way the cascaded stages amplify the signal and overall gain of the amplifier is increased considerably. The output is in phase with i/p.

(4) The total gain is less than the product of the gains of individual stages. This is because when second stage gain follows first, the effective load resistance of first stage is reduced due to shunting effect of i/p resistance of second stage. Thus gain is reduced.

Analyses of RC coupled amplifier



where, $R_b = \frac{R_1 R_2}{R_1 + R_2}$

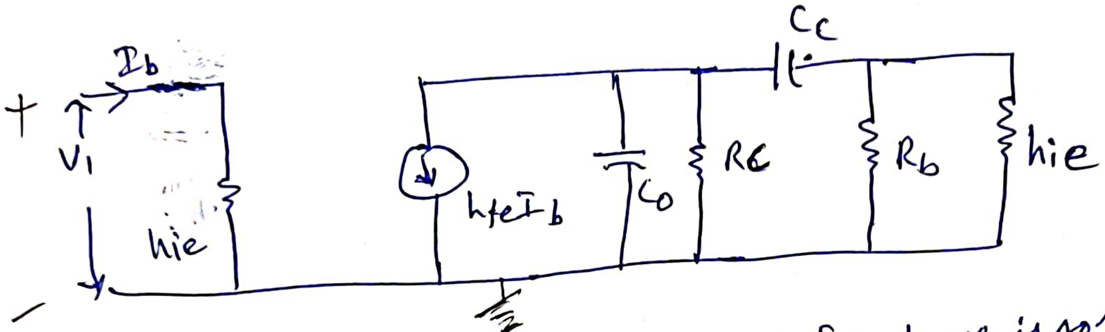
C_O = o/p capacitance of 1st stage

C... (or) i/p capacitance of 2nd stage

Simplified model :- i) h_{re} is small and $h_{re}V_{ce}$ can be neglected

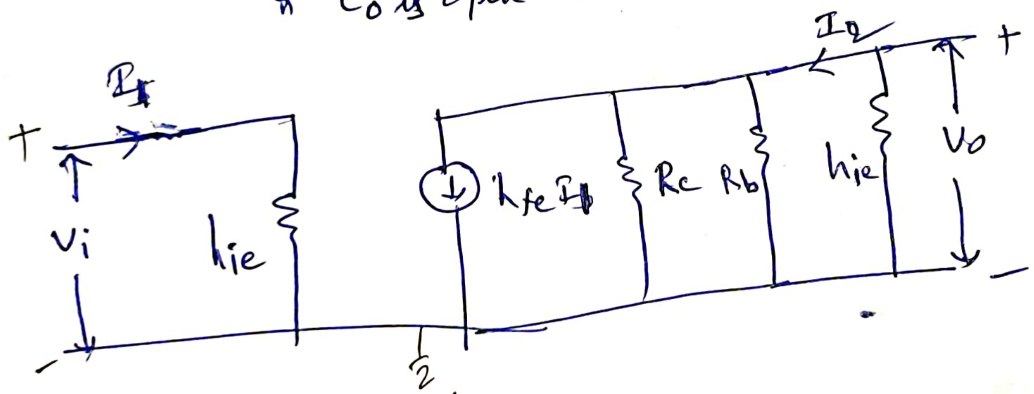
(2) $1/h_{oe}$ is so large and it is considered as open circuit.

(3) The reactance of C_c for any given i/p frequency is so small when compared with parallel combination of R_e and C_o and hence it is effectively considered as short circuit.

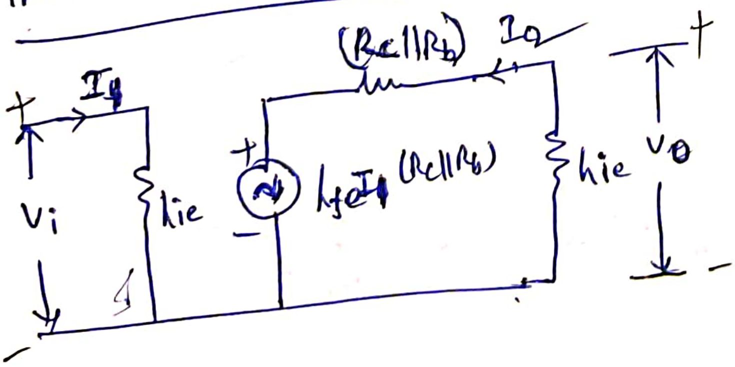


(a) Midfrequency Range :- C_c is short circuit (impedance is so small)

& C_o is open circuit (large impedance)



Thevenin's Equivalent circuit



$$I_2 = \frac{-h_{fe}I_1 \frac{R_cR_b}{R_c+R_b}}{h_{ie}+\frac{R_cR_b}{R_c+R_b}}$$

$$= \frac{-h_{fe}I_1 R_cR_b}{h_{ie}(R_c+R_b)+R_cR_b}$$

Current gain :-

$$\therefore A_{I\,mid} = \frac{I_2}{I_1} = \frac{-h_{fe}R_cR_b}{h_{ie}(R_c+R_b)+R_cR_b}$$

Voltage gain

$$A_V = \frac{V_o}{V_i}$$

$$V_o = I_2 h_{ie} = \frac{-h_{fe} I_1 R_c R_b}{[h_{ie}(R_c+R_b) + R_c R_b]} h_{ie}$$

But, $V_i = I_1 h_{ie}$

$$\therefore V_o = \frac{-h_{fe} R_c R_b}{[h_{ie}(R_c+R_b) + R_c R_b]} \cdot V_i$$

$$\therefore A_{V_{mid}} = \frac{V_o}{V_i} = \frac{-h_{fe} R_c R_b}{h_{ie}(R_c+R_b) + R_c R_b}$$

Advantages :- (1) It is small, light and inexpensive because it requires no expensive (or) bulky components.

(2) It has excellent frequency response and the gains are constant over audio frequency range.

(3) It has minimum non linear distortion because it does not use any coils or transformers.

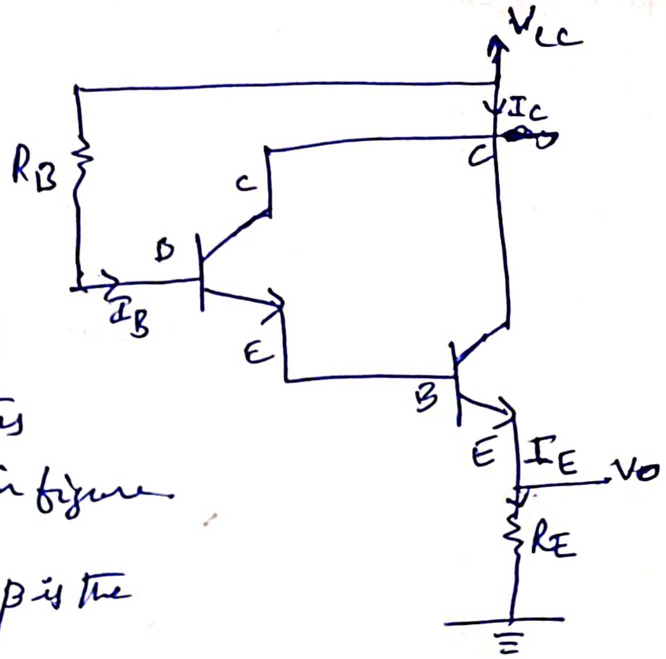
Disadvantages :- (1) The gain of RC coupled amplifier is comparatively small because of loading effect of next stage.

(2) They have a tendency to become noisy.

(3) Impedance matching is poor as o/p impedance is several hundred ohms while of a speaker is only few ohms.

High i/p resistance circuits :-

Darlington Amplifier :-



Darlington circuit of two cascaded emitter followers (CC config) where the second transistor constitutes emitter load for the first as shown in figure.

The current gain is β^2 where β is the gain of individual transistor.

The voltage gain of pair is less than unity as each transistor is connected in emitter follower configuration.

It is also referred as compound amplifier.

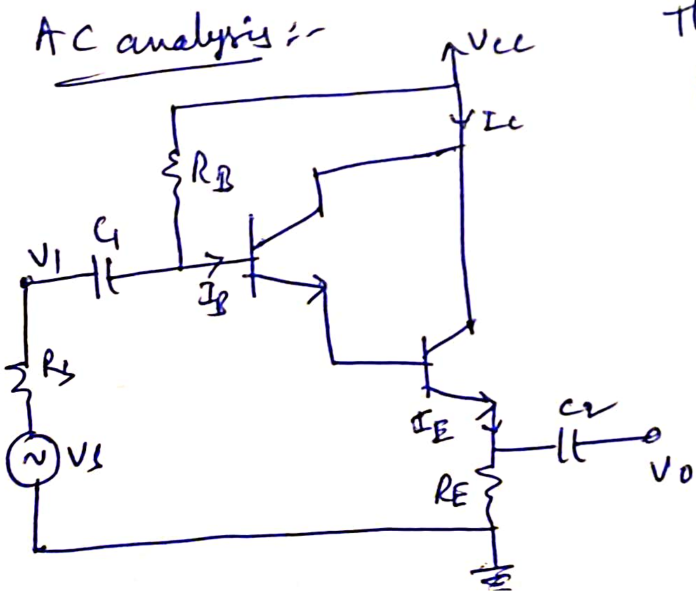
Bias analysis :- Let β_D is the effective β of pair and V_{BE} is the effective base emitter voltage.

Then, $V_{CC} - V_{BE} = I_B R_B + \beta_D I_B R_E$ as $\beta_D >> 1$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta_D R_E}$$

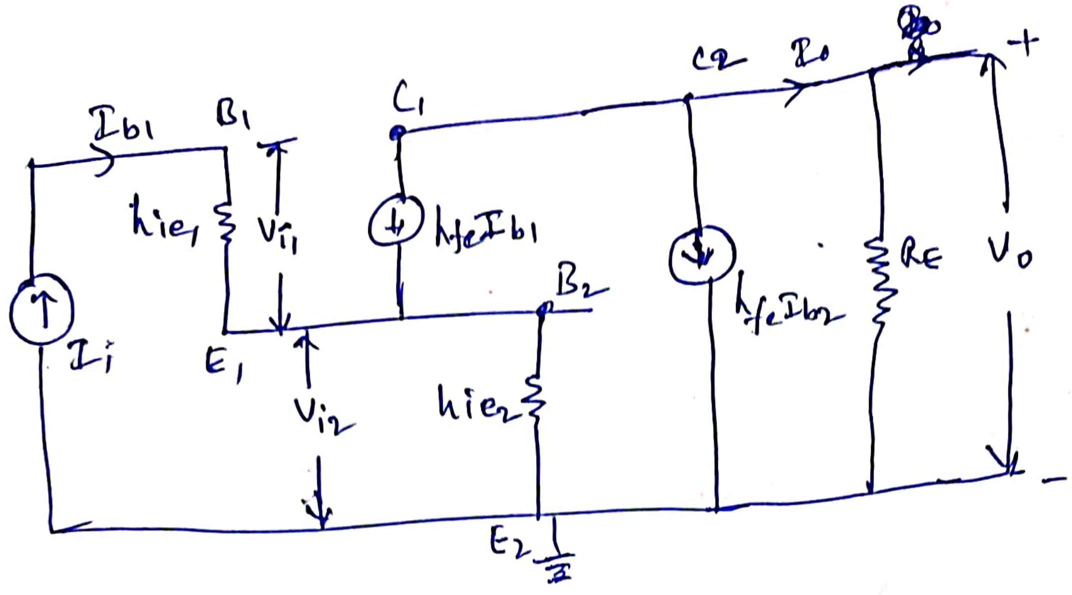
AC analysis :-

The capacitors C_1 and C_2 considered as short in the midband, the ac equivalent circuit is shown in figure below.



(Darlington Emitter follower) ckt.

Equivalent circuit :-



Overall current gain :- $A_I = A_{I_1} \cdot A_{I_2} \simeq \dfrac{(1+h_{fe})^2}{1+h_{oe}h_{fe}R_E} = h_{fe}^2$

I/p Impedance :- $Z_i \simeq \dfrac{(1+h_{fe})^2 R_E}{1+h_{oe}h_{fe}R_E} = h_{fe}^2 R_E$

overall voltage gain :- $A_V \simeq 1 - \dfrac{h_{ie}}{R_{i2}} \simeq 1 - \dfrac{h_{ie}}{(1+h_{fe})R_E} \simeq 1$

Total o/p Impedance :-

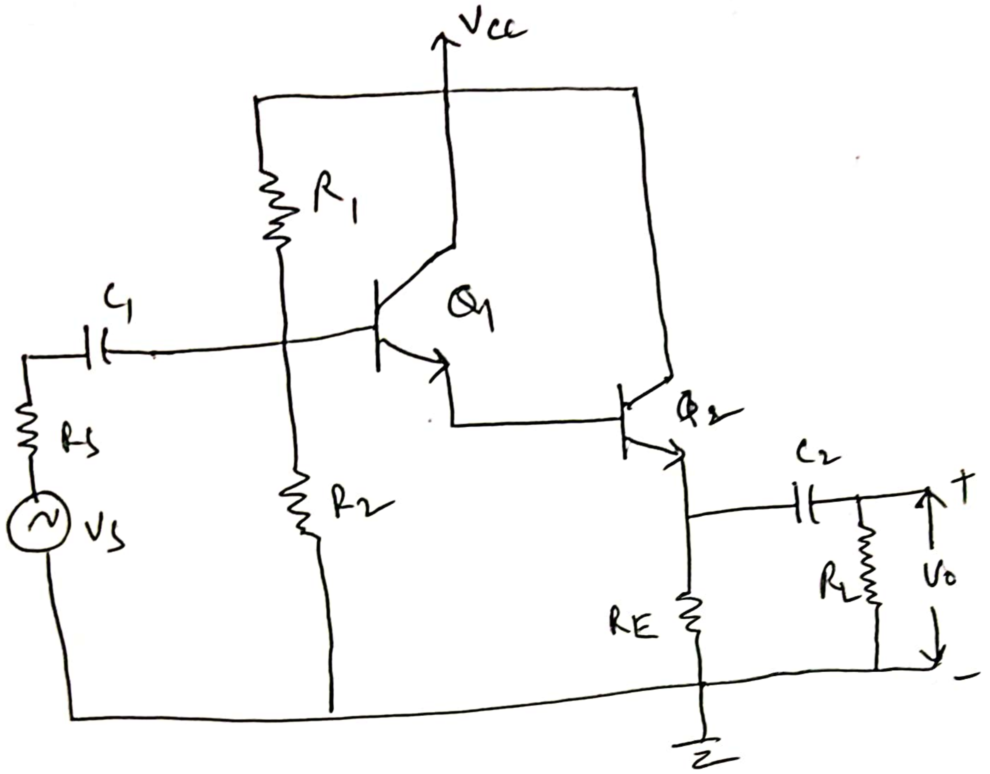
$$Z_{o2} = \frac{R_s + h_{ie}}{(1+h_{fe})^2} + \frac{h_{ie}}{1+h_{fe}}$$

Disadvantages :- (1) overall leakage current may be high.

Merit :- (1) High i/p impedance, but the biasing arrangement reduces Z_i considerable.

Darlington Amplifier

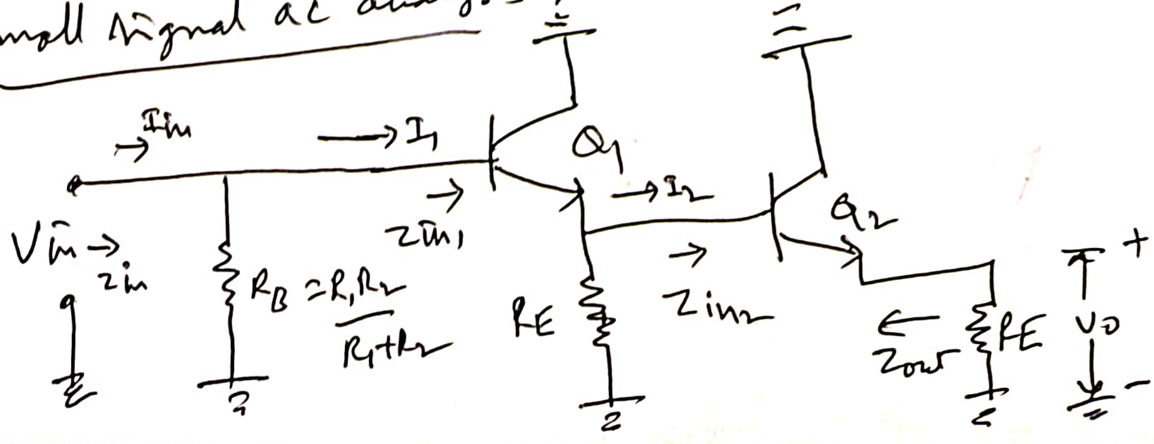
Circuit diagram :-



Darlington amplifier with voltage divider bias is shown in figure above. In this circuit the o/p of one amplifier is coupled into the i/p of next one by directly joining emitter of one transistor to the base of other one as shown in figure. The emitter current of first Transistor becomes the base current for second transistor.

The biasing analysis is similar to that for one transistor except that two V_{BE} drops are to be considered.

Small signal ac analysis :-



(1) current gain :-

For second stage, I/p impedance $Z_{in2} = h_{fe2} R_E$

and current gain, $A_{I_2} = \frac{I_{out}}{I_2} = \frac{I_{e2}}{I_{b2}} = h_{fe2}$

$Z_L = Z_{in2} = h_{fe2} R_E$

$A_{I_1} = \frac{I_2}{I_1} = \frac{I_{e1}}{I_{b1}} = \frac{h_{fe1}}{1 + h_{oe1} h_{fe2} R_E}$

(4) voltage gain

$A_V \cong 1$

$A_I = \frac{I_{out}}{I_1} = A_{I_1} \cdot A_{I_2} = \frac{h_{fe1} \cdot h_{fe2}}{1 + h_{oe1} h_{fe2} R_E}$

For $h_{fe1} = h_{fe2} = h_{fe}$, $h_{oe1} = h_{oe}$

$\therefore A_I = \frac{h_{fe}^2}{1 + h_{oe} h_{fe} R_E} \approx h_{fe}^2 = \beta^2$ $(\because h_{oe} h_{fe} << 1)$

~~$\therefore A_I = \frac{I_{E2}}{I_{b1}} = (1+\beta_2)(1+\beta_1) \cong \beta_1 \beta_2 = \beta^2$ $(\beta_1 = \beta_2 = \beta)$~~

(2) I/p Impedance :- $\because Z_{in2} = h_{fe2} R_E$ is the emitter resistance of first stage

$Z_{in1} = h_{fe1} \left(Z_{in2} \,\|\, \frac{1}{h_{oe1}} \right)$

But, $\frac{1}{h_{oe1}} >> h_{fe2} R_E$

$\therefore Z_{in2} = h_{fe2} R_E$

$Z_{in1} = h_{fe1} \left(h_{fe2} R_E \,\|\, \frac{1}{h_{oe1}} \right) = h_{fe}^2 R_E = \beta^2 R_E$

$(\because h_{fe1} = h_{fe2} = h_{fe})$

$\boxed{\text{I/p impedance is very high}}$

(3) output Impedance :- Z_0 can be determined directly from the emitter equivalent circuits.

$Z_{out1} = \frac{R_s + h_{ie1}}{h_{fe1}}$, $Z_{out2} = \frac{Z_{out1} \,\|\, \frac{1}{h_{oe1}} + h_{ie2}}{h_{fe2}}$

$Z_{out2} << Z_{out1}$ ie, $\boxed{\text{O/p impedance is lowered}}$

5 ⓒ

Applications of Darlington Amplifier

(a) It has enormous Impedance transformation capability i.e, it can transform a low impedance load to a high Impedance load. Hence it is used in high gain operational amplifier which depends on very high i/p impedance for its operation.

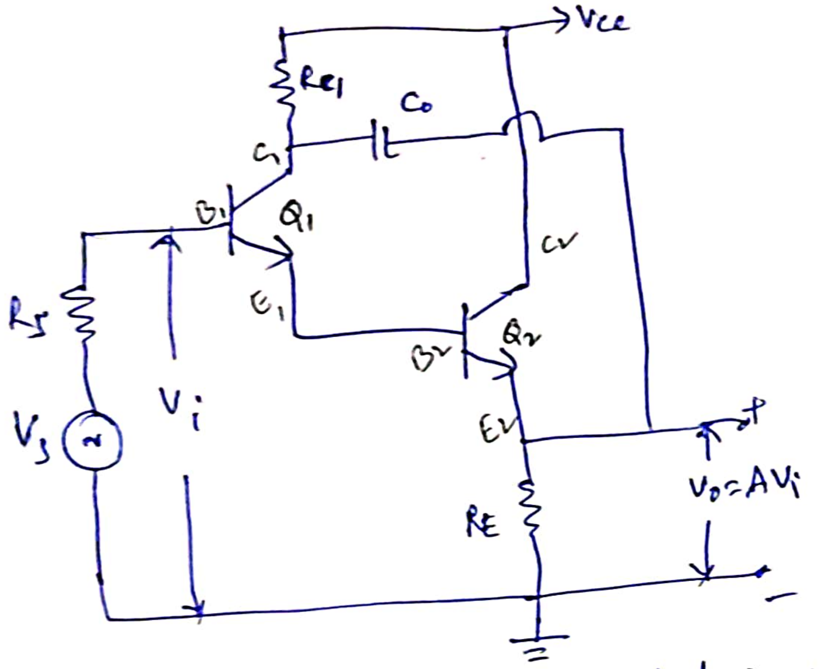
Merits :- (1) uses very few components

(2) High i/p impedance

(3) Low o/p impedance

(4) High current gain

Bootstrapped Darlington Emitter Follower

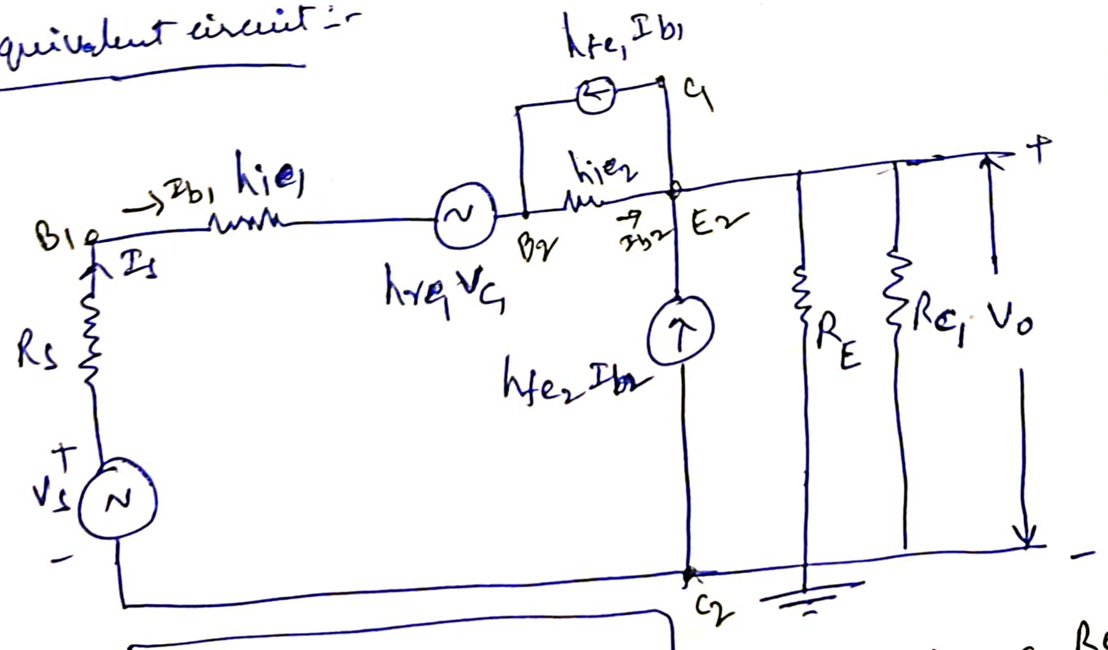


* The i/p resistance of an emitter follower reduced by bias circuit. It may be increased highly by Bootstrapping.
* The Darlington emitter follower has high Zi, but an introduction of biasing resistors R_1 and R_2, the i/p resistance reduced.

By employing Bootstrapping technique to get very high i/p impedance is shown above.

* The circuit is Bootstrapped by inserting C_0 between first transistor collector and second transistor emitter. R_{C_1} is introduced in the collector circuit of first transistor to prevent R_E getting short circuit to ground.

* Equivalent circuit :-



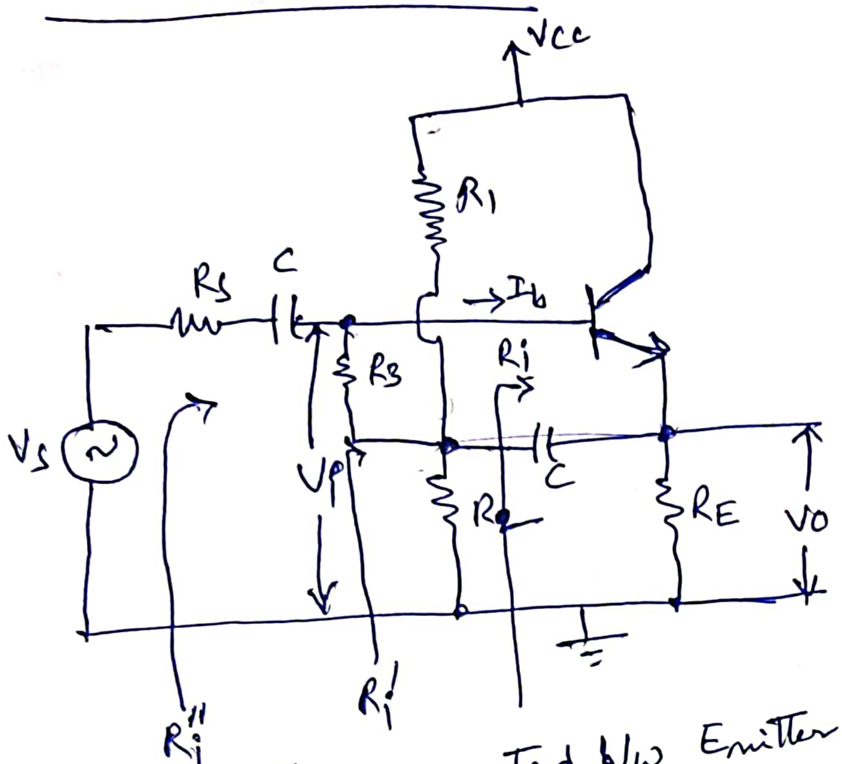
Note :- Max. i/p resistance of a Darlington circuit is limited to 2MΩ

$$\boxed{R_i = \frac{V_i}{I_1} \simeq h_{fe_1} \cdot h_{fe_2} R_{eff}}$$

where $R_{eff} = R_{C_1} \| R_E$

$h_{fe_1} = h_{fe_2} = 100$ & $R_{eff} = 50$ then $R_{in} = 50M\Omega$

Bootstrap Emitter Follower :-



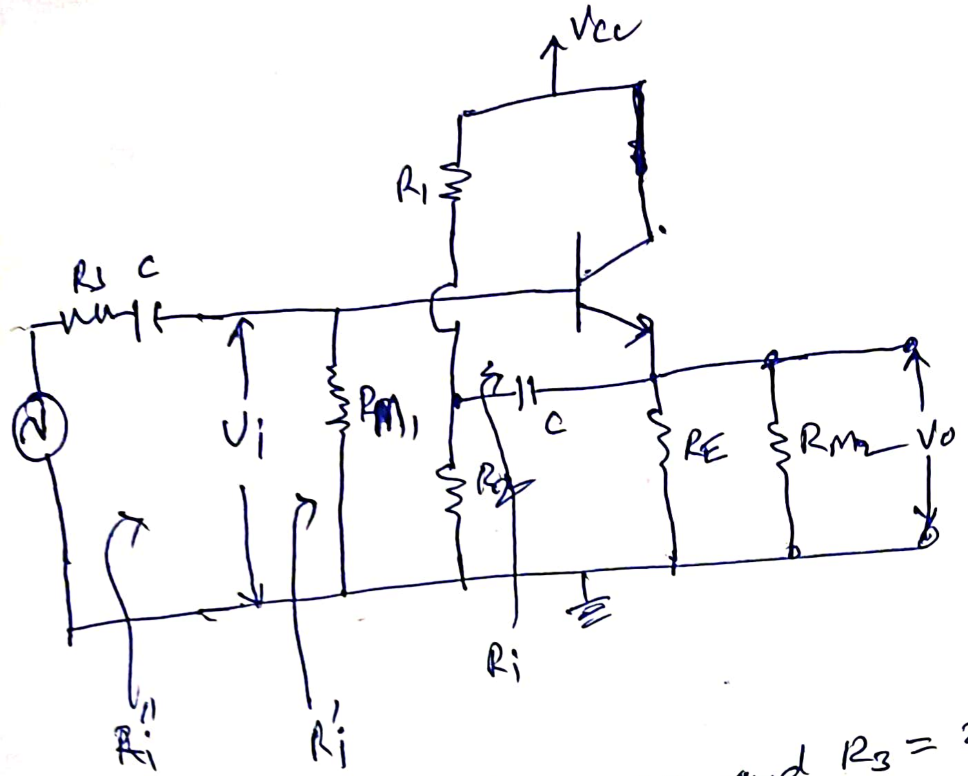
* In Emitter Follower, the i/p resistance of the amplifier is reduced because of shunting effect of biasing resistors.
* To overcome this problem the emitter follower is modified as shown in figure.
* Two additional components R_3 and c are used. The Capacitor 'c' is connected b/w Emitter and junction of R_1, R_2 and R_3.
* For dc signal, 'C' acts as open circuit and therefore R_1, R_2 and R_3 provide necessary biasing to keep transistor in the active region.
* For ac signal, 'C' acts as short circuit. Hence bottom of R_3 effectively connected to o/p (the Emitter), where as the top of R_3 is at its i/p (base) means R_3 connected b/w i/p node and o/p node. For such connection effective i/p resistance is given by Miller's theorem which says that the impedance b/w the two nodes can be resolved into two components, one from each node to ground.

The two components are

$$\frac{Z}{1-K} \text{ and } \frac{ZK}{K-1}.$$

In our case R_3 is the impedance b/w o/p voltage and i/p voltage and K is the voltage gain $\frac{V_o}{V_i} = A_V$.

∴ The Bootstrap circuit with two resolved components of R_3 using Miller's Theorem is shown below.



$$R_{M_1} = \frac{R_3}{1 - A_V}$$

$$R_{M_2} = \frac{R_3 A_V}{A_V - 1}$$

$\because$ for an emitter follower $A_V \approx 1$, R_{M_2} becomes extremely large.

For example $A_V = 0.99$ and $R_3 = 200K$, $\underline{R_{M_2} = 20M}$

Then $R_i' = R_i \,||\, R_{M_1}$

where $R_i = h_{ie} + (1 + h_{fe}) R_E$ for emitter follower.

The above effect $A_V \rightarrow 1$ is called <u>bootstrapping</u>.

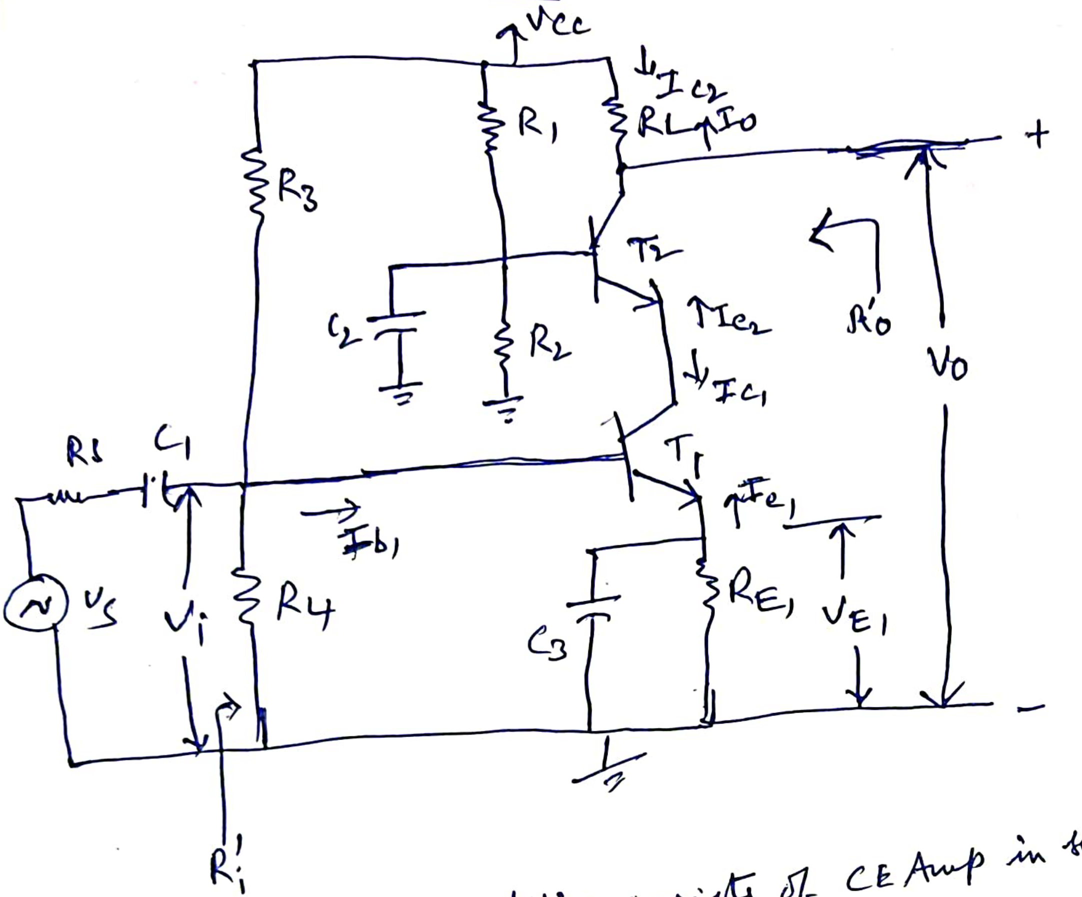
* If one end of R_3 changes in voltage, the other end moves through same potential difference; it is as if R_3 is pulling itself up by its bootstraps.

The effective load on the emitter follower

$$R_{Leff} = R_E || R_1 || R_2 || R_{M_2}$$

$$= \underline{R_E || R_1 || R_2} \quad (\because R_{M_2} \text{ is very large})$$

CE-CB Cascode Amplifier (Cascode Amplifier)



* The cascode amplifier consists of CE Amp in series with a CB Amplifier
* To solve the low impedance problem of a CB circuit.
* T_1 and its associated components operate as a CE amplifier stage, while T_2 functions as a CB o/p stage.
* The cascode Amplifier gives high i/p impedance of CE Amplifier as well as good voltage gain and high frequency performance of a CB circuit.

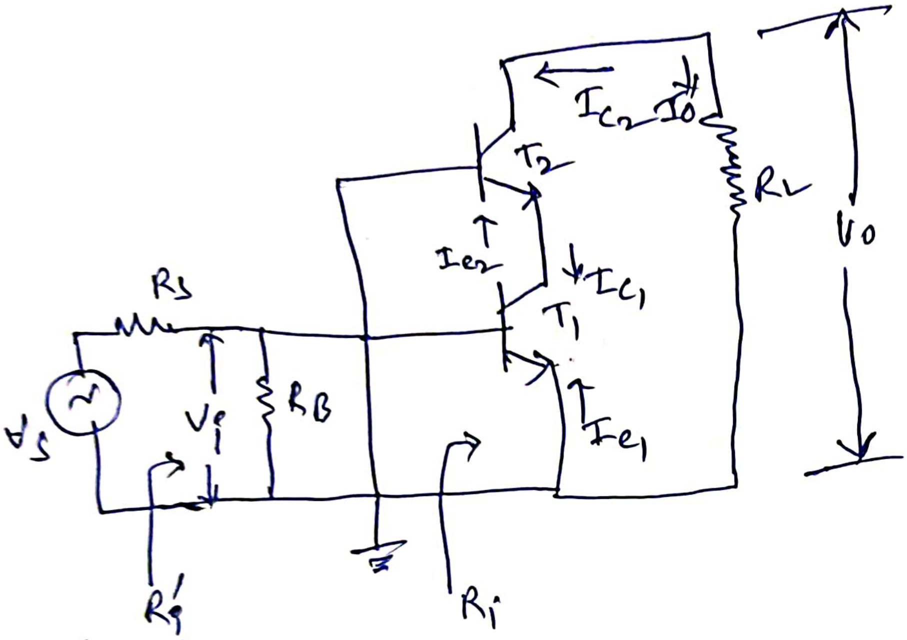
DC bias conditions :- $I_{C1} \approx I_{E1}$, I_{E2} is same as I_{C1}.

$\therefore I_{C2} \approx I_{E1}$

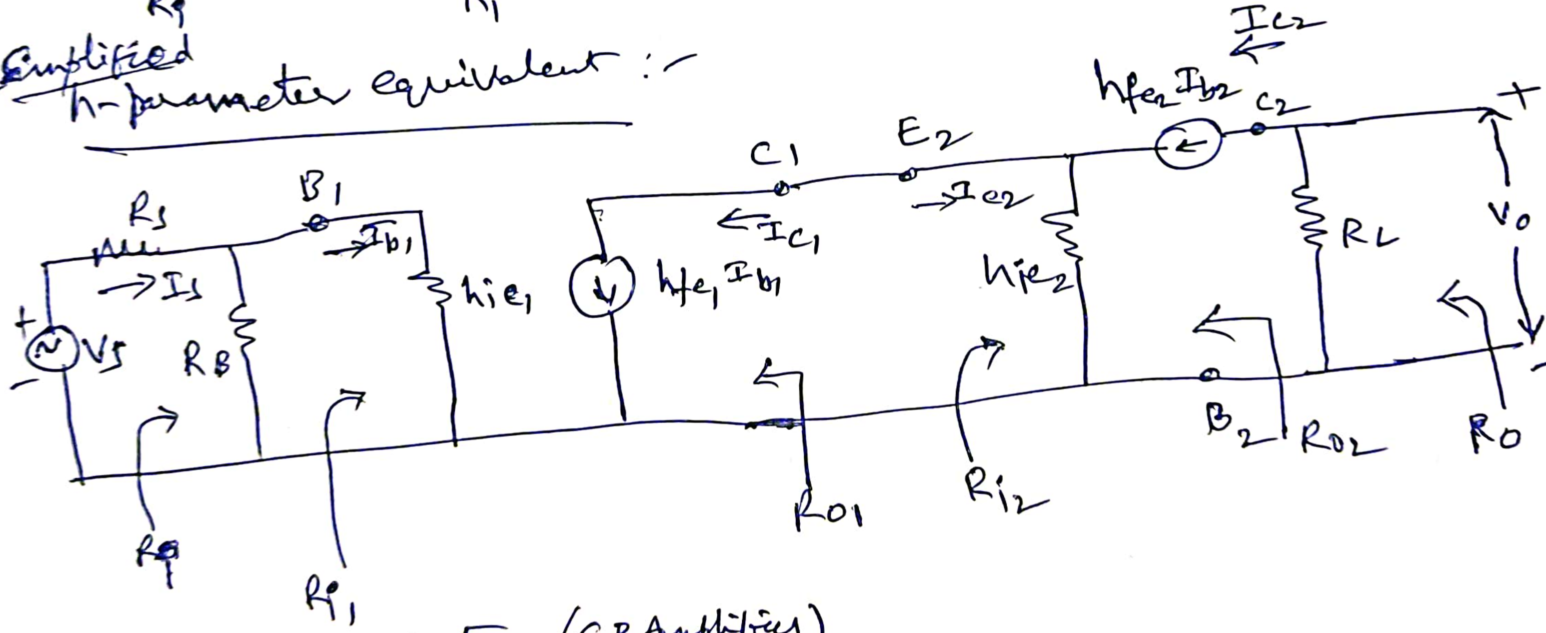
This current remains constant regardless the level of V_{B2} as long as V_{CE1} remains large enough for current operation of T_1.

AC Equivalent circuit :-

$R_B = R_3 || R_4$



Simplified h-parameter equivalent :-



Analysis of second stage (CB Amplifier)

(a) current gain :- $A_{I_2} = \dfrac{h_{fe}}{1+h_{fe}}$

(b) I/p Resistance :- $R_{i_2} = \dfrac{h_{ie}}{1+h_{fe}}$

(c) voltage gain :- $A_{V_2} = \dfrac{A_{I_2} R_{L_2}}{R_{I_2}}$

Analysis of first stage (CE Amplifier)

(a) current gain $(A_{I_1}) = -h_{fe}$

(b) I/p Resistance $(R_{I_1}) = h_{ie}$

(C) Voltage gain $(A_{V_1}) = \frac{A_{I_1} R_{L1}}{R_{I_1}} = \frac{-h_{fe} R_{L1}}{h_{ie}}$

overall Voltage gain :- $A_V = A_{V_1} \times A_{V_2}$

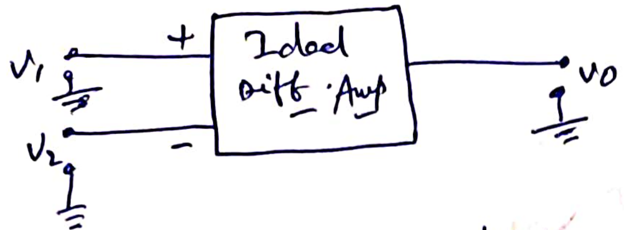
overall i/p resistance $R_i = R_{I_1} || R_B = R_{I_1} || R_3 || R_4$

output resistance (R_O) : $R_{O1} = \infty$

$R_{O2} = \infty$

$R_O = R_{O2} || R_L = R_L$

Basics Of Difference Amplifier



* Differential Amplifier amplifies the difference b/w two i/p signals

$$V_o \propto (V_1 - V_2)$$

Differential gain (A_d)

$$V_o = A_d (V_1 - V_2)$$

Differential gain $= A_d =$ gain with which differential amplifier amplifies the difference b/w two signals.

$$V_o = A_d \cdot V_d \Rightarrow A_d = \frac{V_o}{V_d}$$

$$A_d \text{ in dB} = 20 \log_{10}(A_d) \text{ in dB}$$

Common mode gain (A_c)

(in case of practical diff. Amp)

$$V_c = \frac{V_1 + V_2}{2}, \quad V_o = A_c \cdot V_c$$

$$A_c = V_o / V_c$$

∴ Total o/p of any differential Amp is given by

$$\boxed{V_o = A_d V_d + A_c V_c}$$

CMRR (Common Mode Rejection Ratio) :-

$$CMRR = \rho = \left| \frac{A_d}{A_c} \right|$$

The ability of differential amplifier to reject the common mode signal is CMRR.

Ideally, $A_c = 0$, then $CMRR = \infty$

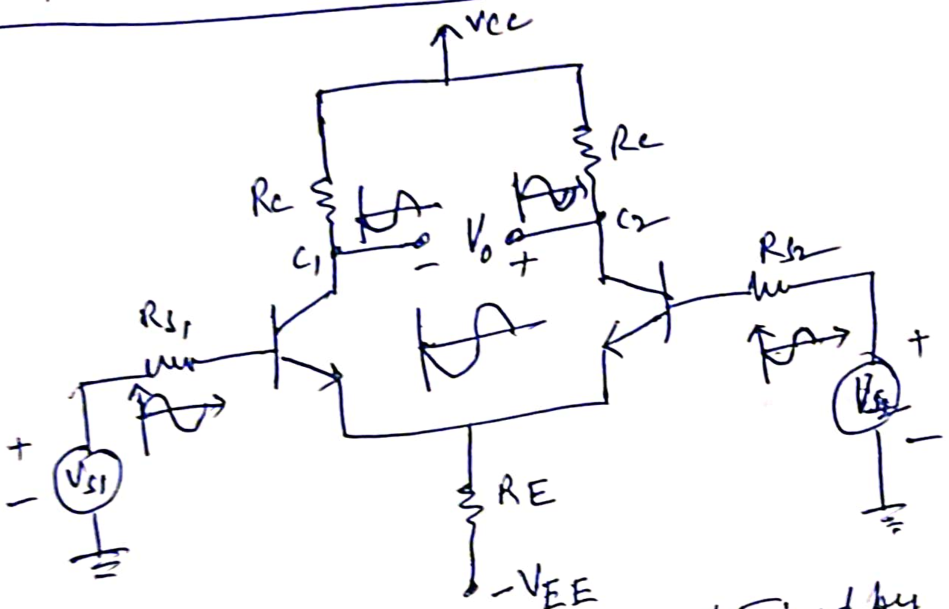
practically, A_d is large
A_c is small
∴ CMRR = very large

Features of Differential Amplifier

(1) High A_d
(2) Low A_c
(3) High CMRR
(4) Two i/p terminals.
(5) High Z_i
(6) Large BW
(7) Low Z_o

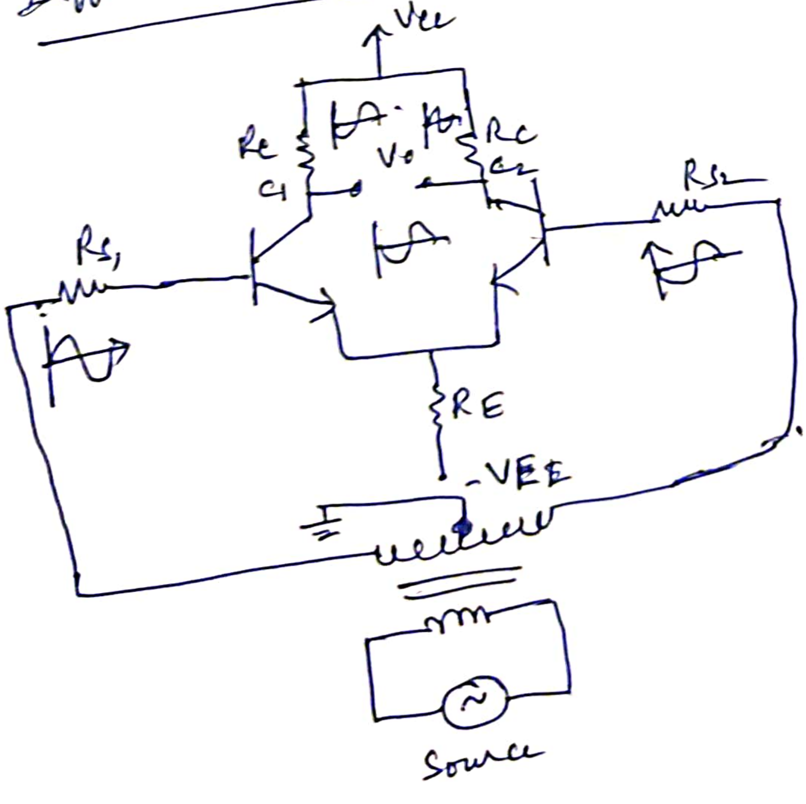
Dual i/p, balanced o/p differential amplifier (Emitter coupled Diff Amp)

$V_{CC} = -V_{EE}$



* Differential Amplifier can be obtained by two emitter biased ckts.

Differential mode operation:

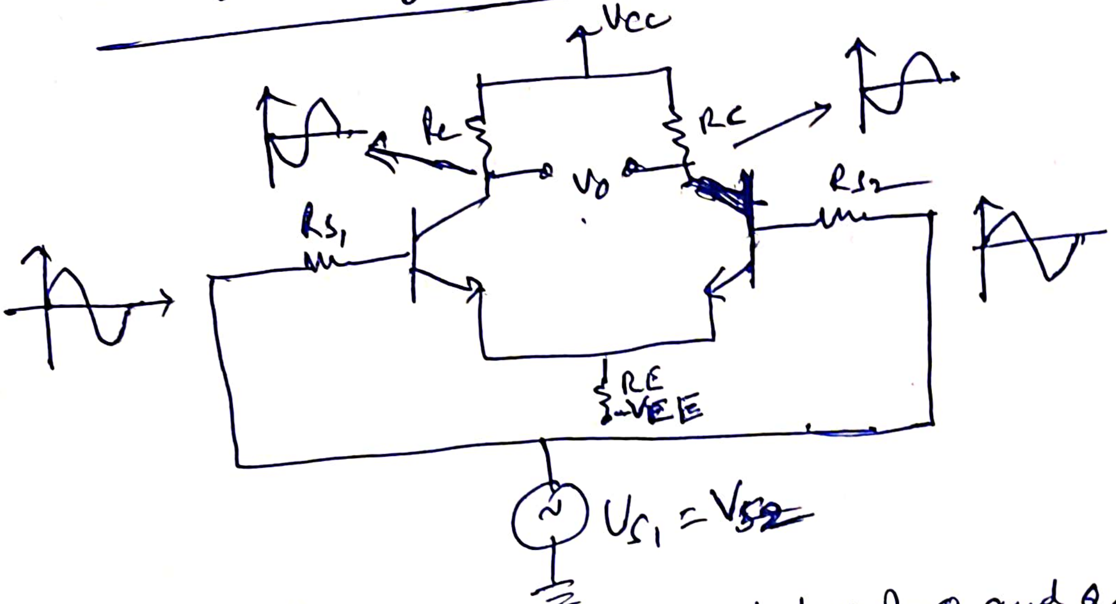


* Same i/p's but out of phase are obtained from centre tapped transformer.
* No signal across R_E and does not introduce negative feedback

$\therefore V_0 = V_m - (-V_m) = 2V_m$

$\therefore \boxed{V_0 = 2V_m}$

Common mode ~~gain~~ operation :-



$V_o = 0$

V_{S_1} & V_{S_2} are in same phase

The phase signal voltages at bases of Q_1 and Q_2 causes in phase signal voltages to appear across R_E, which add together. Hence R_E carries a signal current and provides a negative feedback. This FB reduces the common mode gain of differential amplifier.

Ideally it should be zero.

$V_o = V_1 - V_2 = 0$

Comparison b/w Various Cascading Methods

Parameter	RC coupled	Transformer coupled	Direct coupled
Coupling components	R and C	Impedance matching Transformer	—
Block DC	Yes	Yes	No
Frequency Response	Flat at middle frequencies	Not uniform, high at resonant frequency and low at other frequency	Flat at middle frequencies and improvement in low frequency response
Impedance matching	Not achieved	achieved	Not achieved
DC Amplification	No	No	Yes
Weight	Light	Bulky and heavy	—
Drift	Not present	Not present	present
Application	Used in all audio small signal amplifiers. Used in record players, tape recorders, PA systems, radio receivers, TV receivers	Used in amplifiers where impedance matching is important. Used in o/p stage of PA system to match the impedance of loud speaker. Used in RF amp stage of receiver as a tuned amplifier	Used in DC amp is required.